

3.7 Nitrogen Compounds

Question Paper

Course	CIEA Level Chemistry
Section	3. Organic Chemistry
Topic	3.7 Nitrogen Compounds
Difficulty	Medium

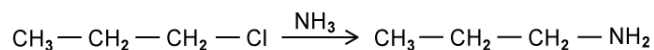
Time allowed: 70

Score: /56

Percentage: /100

Question 1a

Propylamine is a colourless liquid that is used to manufacture a variety of chemicals including textile resins, drugs and pesticides. It can be produced by the reaction of 1-chloropropane with ammonia.



i)

State the role of the ammonia in this reaction.

[1]

ii)

Name the type of reaction that occurs in this synthesis of propylamine.

[1]

[2 marks]

Question 1b

Use an appropriate chemical equation to explain why propylamine can be described as a Brønsted-Lowry base with water.

[3 marks]

Question 1c

The reaction between 1-chloropropane and ammonia proceeds in two steps.

Step 1 involves the formation of an intermediate.

Step 2 forms the final propylamine product.

As the reaction proceeds, the bond angles around the nitrogen atom change. Explain these changes in the bond angles around the nitrogen atom.

[3 marks]

Question 2a

Fig. 1.1 shows a reaction sequence to form an amine.

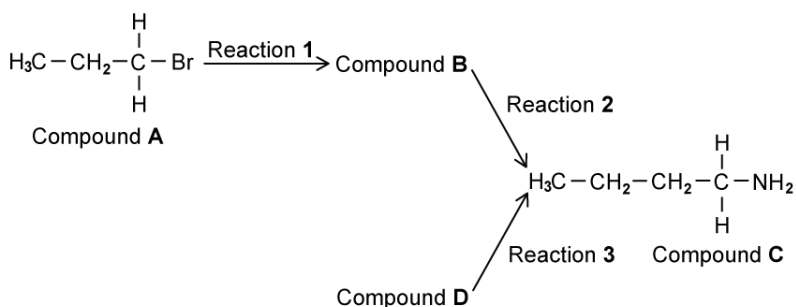


Fig. 1.1

i)
Name compound **A**.

[1]

ii)
Reaction 1 heats compound **A** under reflux with an ethanolic solution of potassium cyanide.

Draw the mechanism for the reaction of compound **A** with the ethanolic solution of potassium cyanide to form compound **B**.

- Identify the ion that reacts with compound **A**
- Draw the product of this reaction, compound **B**.
- Include all charges, partial charges, lone pairs and curly arrows.

[4]

[5 marks]

Question 2b

In reaction **2**, vapours of compound **B** are converted into compound **C** by reacting them with hydrogen gas over a nickel catalyst.

Name the type of reaction that occurs in reaction **2**.

[1 mark]

Question 2c

Compound **C** can also be produced in a single step from compound **D**.

i)

Draw the displayed formula of compound **D**.

[1]

ii)

State suitable reagents and conditions for reaction **3**.

[2]

[3 marks]

Question 2d

Compound **C** is a primary amine as there is one alkyl chain attached to the nitrogen atom.

Draw the skeletal formula of compound **C**'s two isomers, which are also primary amines.

[2 marks]

Question 3a

Propanal undergoes nucleophilic addition with a mixture of HCN and NaCN to make 2-hydroxybutanenitrile, $\text{CH}_3\text{CH}_2\text{CH}(\text{OH})\text{CN}$.

The mechanism for this reaction occurs in two main steps. The first step involves a nucleophile attacking the carbonyl carbon of propanal.

Explain which species acts as the nucleophile during this reaction.

[2 marks]

Question 3b

The second step in the mechanism involves an intermediate species, $\text{CH}_3\text{CH}_2\text{C}(\text{O})\text{HCN}^-$, reacting with HCN to form 2-hydroxybutanenitrile.

Draw the mechanism for this step.

- Identify the intermediate species that reacts with HCN.
- Include all charges, partial charges, lone pairs and curly arrows.

[3 marks]

Question 3c

The structure of the 2-hydroxybutanenitrile product is shown in Fig. 1.1.

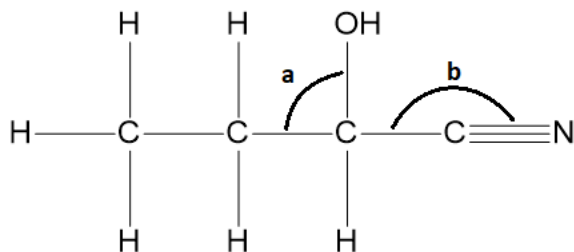


Fig. 1.1

Predict the values for the bond angles **a** and **b** shown in the diagram.

[2 marks]

Question 3d

The product 2-hydroxybutanenitrile exists as a pair of stereoisomers.

i)

Name the type of stereoisomerism shown by 2-hydroxybutanenitrile.

[1]

ii)

Draw three-dimensional diagrams of this pair of stereoisomers. Indicate with an asterisk (*) the chiral centre on one of the structures drawn.

[3]

[4 marks]

Question 4a

Two students try to prepare butanoic acid from a halogenoalkane in the laboratory.

Both students perform a nucleophilic substitution reaction to form **P** in reaction 1 of **Fig. 1.1**.

They both form butanoic acid from **P** but student **A** uses a single reaction, while student **B** uses two reactions.

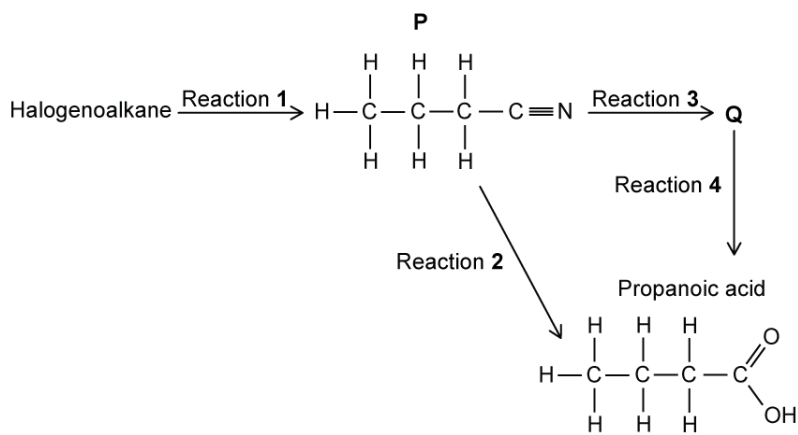


Fig. 1.1

- i)
State the reagents and conditions required for reaction 1.

[2]

- ii)
Student **A** uses 1-bromopropane to form **P**.

Student **B** uses a different halogenoalkane which forms **P** at a slower rate.

Suggest the identity of the halogenoalkane that student **B** uses. Explain your answer.

[2]

[4 marks]

Question 4b

Student **A** converts **P** into butanoic acid in a single step, reaction 2.

i)

Name the type of reaction occurring in reaction 2.

[1]

ii)

State an appropriate reagent for reaction 2 and name the other product of the reaction.

[1]

[2 marks]

Question 4c

Student **B** converts **P** into **Q** using dilute sodium hydroxide in reaction 3. This is followed by acidification of the product in reaction 4 to form butanoic acid.

Draw the structural formula of **Q**.

[1 mark]

Question 4d

Explain why it is likely that student **A** will have a higher overall yield of butanoic acid from their halogenoalkane than student **B**.

[2 marks]

Question 5a

Halogenoalkanes are often used as intermediates in organic reactions.

Three reactions of bromoethane, $\text{CH}_3\text{CH}_2\text{Br}$, are shown in Fig. 1.1.

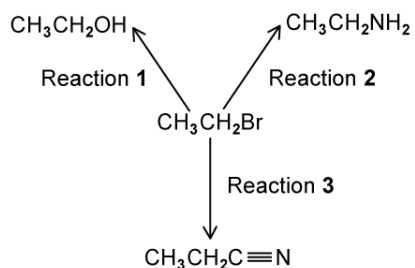


Fig. 1.1

For each reaction, state the reagent and solvent used.

reaction 1	reagent
	solvent
reaction 2	reagent
	solvent
reaction 3	reagent
	solvent

[6 marks]

Question 5b

Reactions 1 and 3 require specific reaction conditions to occur as shown in **Fig. 1.1**.

For each reaction, give another solvent and the alternative product that will be formed.

reaction 1	alternative solvent
	alternative product
reaction 3	alternative solvent
	alternative product

[4 marks]

Question 5c

The product of reaction 2 can be converted into CH_3CN .

i)

Name the compound CH_3CN .

[1]

ii)

Name the type of reaction used to form CH_3CN .

[1]

[2 marks]

Question 5d

The product of reaction 3 can be used to produce propanoic acid by two different hydrolysis reactions.

Compare the two types of hydrolysis reaction in the production of propanoic acid from the product of reaction 3.

[5 marks]

Model Answers: Medium

1a

a)

i) The role of the ammonia in this reaction is:

- A nucleophile; [1 mark]

ii) The type of reaction that occurs in this synthesis of propylamine is:

- (Nucleophilic) substitution; [1 mark]

[Total: 2 marks]

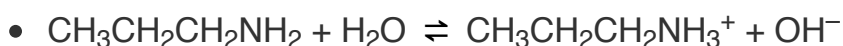
- The lone pair of the ammonia molecule attacks the δ^+ carbon of the C-Cl bond
 - This makes the ammonia a nucleophile
- The Cl group is replaced by an NH_2 group
 - This is substitution

1b

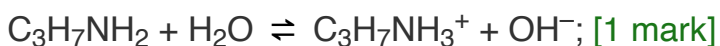
b) Propylamine can be described as a Brønsted-Lowry base with water because:

- It / propylamine has a lone pair of electrons; [1 mark]
- Which can accept a proton / H^+ ; [1 mark]

Equation:



OR



[Total: 3 marks]

- **Remember:** A Brønsted-Lowry base is a species that accepts a proton /

H⁺ through a lone pair of electrons

- **Careful:** The question asks you to explain
 - This means that stating it "*accepts a proton / H⁺*" will not be sufficient for the mark as this is a description of how it behaves as a Brønsted-Lowry base
 - The explanation must include a comment about the lone pair of electrons
- The equation must show the propylamine taking / accepting a proton from water
 - This will form the CH₃CH₂CH₂NH₃⁺ ion and a hydroxide ion / OH⁻

1c

c) To explain these changes in the bond angles around the nitrogen atom:

- The bond angle around the nitrogen atom in ammonia **AND** in propylamine is 107°
AND
The bond angle around the nitrogen atom in the intermediate is 109.5°; [1 mark]
- The nitrogen lone pair in ammonia and propylamine cause greater / more repulsion
AND
Which causes the molecules to have a pyramidal shape; [1 mark]
- There are four / all bonding pairs around the nitrogen atom in the intermediate
AND
Which causes it to have a tetrahedral shape; [1 mark]

[Total: 3 marks]

- **Remember:** Bond angles and shapes of molecules are connected
- Focusing on the nitrogen atom
 - In ammonia
 - There is one lone pair and three bonding pairs
 - Lone pairs cause greater repulsion than bonding pairs
 - This causes ammonia to have a pyramidal shape with a bond angle of 107°
 - In the intermediate, there are only bonding pairs around the nitrogen atom
 - This means that the intermediate has a tetrahedral shape with a bond angle of 109.5°
 - In propylamine, there is one lone pair and three bonding pairs around the nitrogen atom
 - This causes propylamine to be pyramidal shaped around the nitrogen atom, with a bond angle of 107°

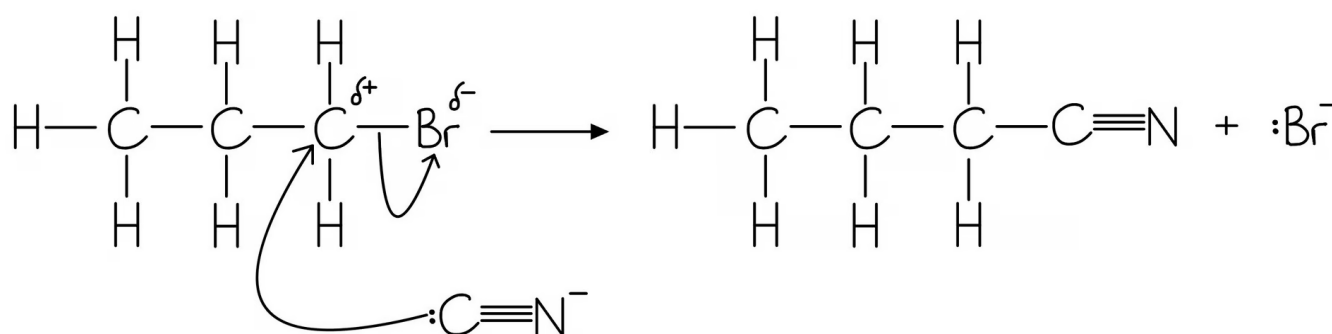
2a

a)

i) Compound **A** is:

- 1-bromopropane; [1 mark]

ii) The mechanism for the reaction of compound **A** with the ethanolic solution of potassium cyanide to form compound **B** is:



- $:\text{CN}^-$

OR

- $:\text{C}\equiv\text{N}^-$; [1 mark]

- Curly arrow from the lone pair on the $:\text{CN}^-$ ion to the δ^+ carbonyl carbon; [1 mark]
- Correct dipole on the C-Br bond

AND

Curly arrow from the C-Br bond to the bromine; [1 mark]

Correct product shown for compound **B**; [1 mark]

[Total: 5 marks]

- For part (i)
 - Compound **A** has three carbons with single bonds, so the name is propane based
 - It has a bromine atom on carbon 1, so the name features 1-bromo
 - This gives 1-bromopropane
- For part (ii)
 - The reaction with an ethanolic solution of potassium cyanide will involve the cyanide ion / $:\text{CN}^-$

- The negative charge is actually on the carbon atom
- It is accepted to have the negative charge associated with the whole ion
- Compound **A** has a dipole on the C-Br bond with the carbon being $\delta+$ and the bromine being $\delta-$
- This means that the $:\text{CN}^-$ ion, which is a nucleophile, will attack the $\delta+$ carbon
- This results in the C-Br bond breaking and both electrons from that bond moving to the bromine atom forming a bromide ion / Br^- and butanenitrile as the organic product

2b

b) The type of reaction that occurs in reaction **2** is:

- Reduction; [1 mark]

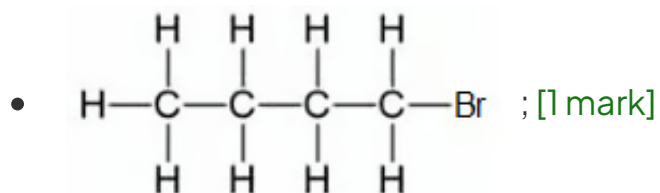
[Total: 1 mark]

- In reaction **2**, nitrile compound **B** gains hydrogen to form compound **C**.
- It is more common to talk about reduction and oxidation in terms of oxygen or electrons, but gaining hydrogen is also reduction and losing hydrogen is oxidation

2c

c)

i) The displayed formula of compound **D** is:



ii) Suitable reagents and conditions for reaction **3** are:

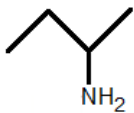
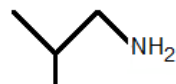
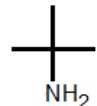
- Heated under pressure; [1 mark]

- The reaction to form an amine directly is the nucleophilic substitution reaction of a halogenoalkane with ethanolic ammonia
 - This occurs while being heated under pressure
- The most common compound seen in exams for compound **D** is 1-bromobutane
 - 1-chlorobutane is also accepted

2d

d) The skeletal formula of two isomers of compound **C**, which are also primary amines are:

Any **two** of the following:

-  ; [1 mark]
-  ; [1 mark]
-  ; [1 mark]

[Total: 2 marks]

- Compound **C** is butylamine, which consists of a butane chain with an amine group attached to carbon-1
- One isomer would be to shift the amine group from carbon-1 to carbon-2
- A second isomer comes from moving the end methyl / CH₃ group and placing it on carbon-2 resulting in the branched isomer 2-methylpropylamine
- A third isomer comes from starting with a propane chain, adding an amine group and a methyl group **both** on carbon-2

3a

a) The species which acts as the nucleophile during this reaction is:

- CN⁻ / :CN⁻ / the cyanide ion; [1 mark]

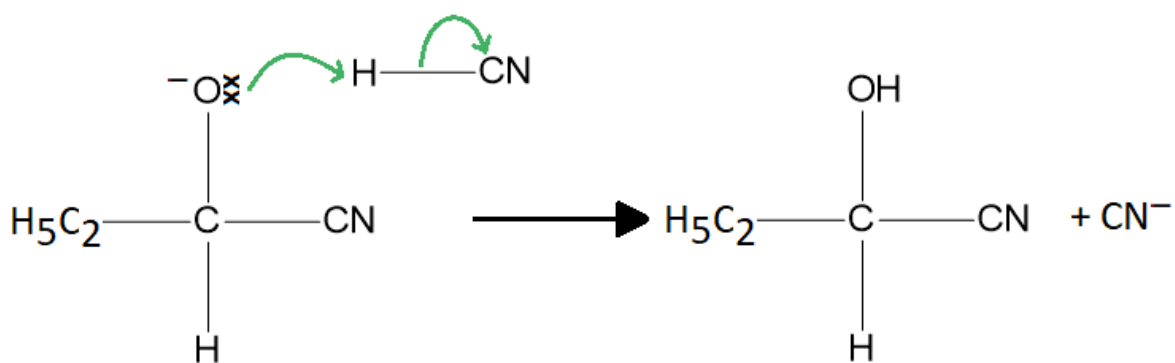
- (Because,) it has a lone pair / is a lone pair donor; [1 mark]

[Total: 2 marks]

- **Remember:** A nucleophile is a species that is attracted to areas of (partial) positive charge
- This means that they are an electron-rich species that is able to donate a pair of electrons
- The NaCN is made from an Na^+ ion and the CN^- ion
- The CN^- ion has an available lone pair which makes it the nucleophile in this reaction

3b

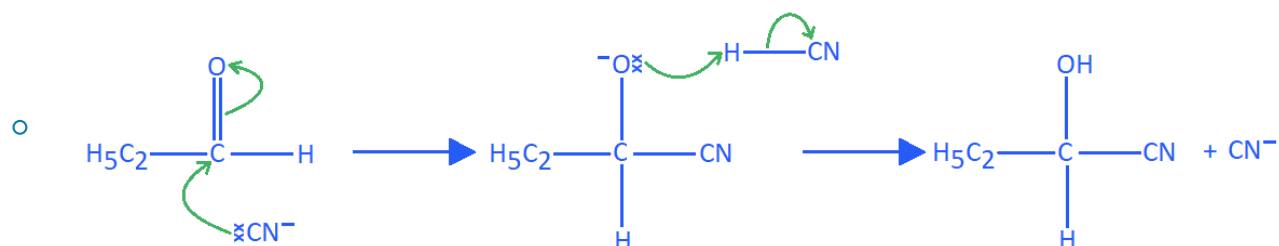
b) The mechanism for the second step involving the intermediate species reacting with HCN to form 2-hydroxybutanenitrile is:



- Correct intermediate drawn including negative charge **AND** lone pair on the oxygen atom; [1 mark]
- Curly arrow from the oxygen lone pair to the hydrogen atom (of HCN); [1 mark]
- Curly arrow from the H-C bond (of HCN) to the carbon atom; [1 mark]

[Total: 3 marks]

- The complete mechanism for this reaction using the nucleophile identified in part (a) is:



- Part (a) worked with the cyanide ion nucleophile
 - The negative charge is actually on the carbon atom
 - It is accepted to have the negative charge associated with the whole ion
- So, this question focuses on the second half of the mechanism and, therefore, requires a sound understanding of the mechanism

3c

c) The bond angles **a** and **b** are:

- **a** = 109.5 (°); [1 mark]
- **b** = 180 (°); [1 mark]

[Total: 2 marks]

- The bond shown in **a** is one of four bonding pairs of electrons
 - This means that the molecule will adopt a tetrahedral shape around that carbon resulting in bond angles of 109.5 °
- The bond shown in **b** is also one of four bonding pairs of electrons
 - However, the C ≡ N means that bond **b** is associated with a linear shape about that carbon, which has a bond angle of 180 °

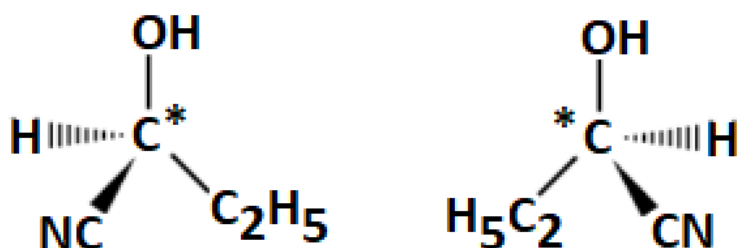
3d

d)

i) The type of stereoisomerism shown by 2-hydroxybutanenitrile is:

- Optical (isomerism); [1 mark]

ii) One example of the three-dimensional diagrams of this pair of stereoisomers are:



- One 3-D structure correct; [1 mark]

- The mirror image of the molecule drawn for the first mark

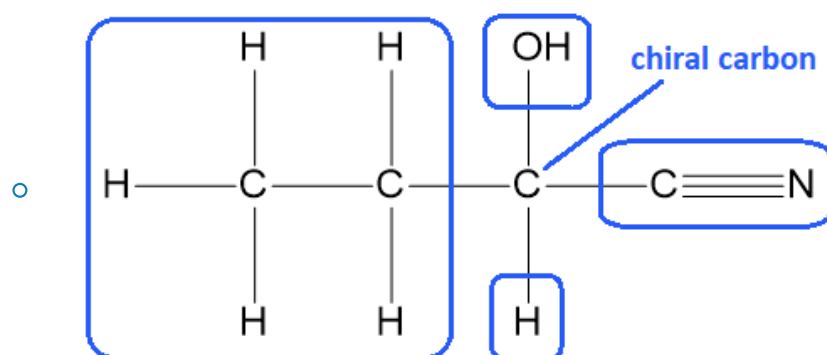
OR

The same diagram with two groups swapped; [1 mark]

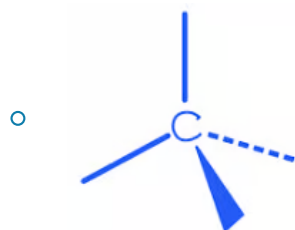
- The chiral carbon correctly labelled on one of the diagrams; [1 mark]

[Total: 4 marks]

- The chiral carbon needs to be identified first:
 - **Remember:** A chiral carbon has four different atoms or groups of atoms around it



- This shows that the molecule has an OH group, CN group, H atom and C₂H₅ group
- You can then apply these to a chiral carbon 3-D structure to get one of the optical isomers and then draw the mirror image for the second structure

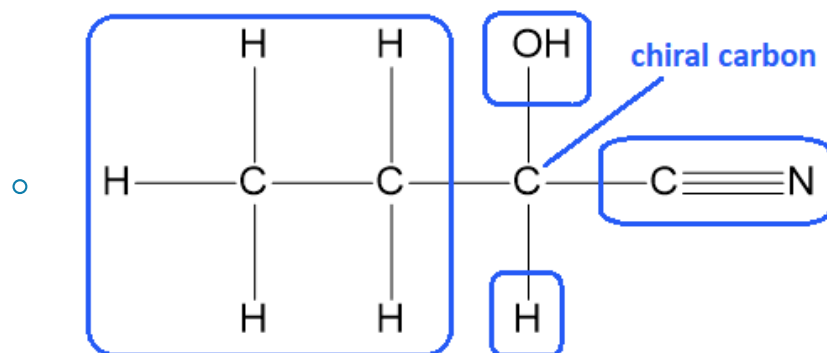


4a

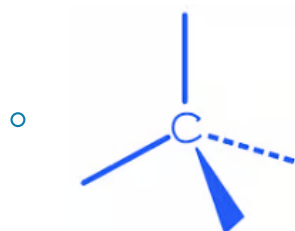
a)

i) The reagents and conditions required for reaction 1 are:

- Heated under reflux; [1 mark]



- This shows that the molecule has an OH group, CN group, H atom and C₂H₅ group
- You can then apply these to a chiral carbon 3-D structure to get one of the optical isomers and then draw the mirror image for the second structure



4a

a)

i) The reagents and conditions required for reaction 1 are:

- Heated under reflux; [1 mark]
- With a solution of sodium cyanide / NaCN in ethanol

OR

With a solution of potassium cyanide / KCN in ethanol; [1 mark]

ii) The identity of the halogenoalkane that student **B** uses is:

- 1-chloropropane

OR

1-fluoropropane; [1 mark]

- (Because,) the carbon-chlorine / C-Cl bond is stronger than the carbon-bromine / C-Br bond

OR

(Because,) the carbon-fluorine / C-F bond is stronger than the carbon-bromine / C-Br

bond

OR

(Because,) the reactivity of the halogenoalkanes increases as you move down the halogens

OR

(Because,) the halogenoalkanes above bromopropane are less reactive / react more slowly; [1 mark]

[Total: 4 marks]

- You are expected to know the reagents and conditions required to convert a halogenoalkane into a nitrile
 - You must state that the sodium cyanide (or potassium cyanide) is in ethanol / alcoholic conditions
 - If the sodium cyanide (or potassium cyanide) is in water, then a different reaction takes place
 - If you do not state "in ethanol" then the examiners will assume that you mean an aqueous solution and you will lose a mark
- **Remember:** The reactivity of the halogenoalkanes increases as you move down the halogens
 - This is because the carbon-halogen bond is weaker as you move down the group as the halogen is less electronegative

4b

b)

i) The type of reaction occurring in reaction 2 is:

- Acid hydrolysis; [1 mark]

ii) An appropriate reagent for reaction 2 and the other product of the reaction are:

Any **one** of the following:

- Hydrochloric acid

AND

Ammonium chloride; [1 mark]

- Sulfuric acid
AND
Ammonium sulfate; [1 mark]
- Nitric acid
AND
Ammonium nitrate; [1 mark]

[Total: 2 marks]

- It is important that you know how the two different hydrolysis reactions can result in the same carboxylic acid
 - Reaction 2 is acid hydrolysis to form the carboxylic acid
 - This results in the formation of the carboxylic acid **AND** an ammonium salt
 - The ammonium salt is dependent on the acid used
 - Reaction 3 is the alkaline hydrolysis followed by reaction 4, which is acidification
 - The combination of reactions 3 and 4 will form the carboxylic acid

4c

c) The structural formula of **Q** is:

- $\text{CH}_3\text{CH}_2\text{CH}_2\text{COONa}$
OR
 $\text{CH}_3(\text{CH}_2)_2\text{COONa}$
OR
 $\text{CH}_3\text{CH}_2\text{CH}_2\text{COO}^- \text{Na}^+$
OR
 $\text{CH}_3(\text{CH}_2)_2\text{COO}^- \text{Na}^+$; [1 mark]

[Total: 1 mark]

- Alkaline hydrolysis of a nitrile forms the carboxylate salt **AND** ammonia
- The carboxylate salt is then acidified to form the carboxylic acid

4d

d) It is likely that student **A** will have a higher overall yield of butanoic acid from their halogenoalkane than student **B** because:

- Student **A** completes the preparation / process in two steps
AND
Student **B** completes the preparation / process in three steps; [1 mark]
- Some product will be lost during each step
AND
So, student **A** will lose less product overall / student **B** will lose more product overall (resulting in a lower yield); [1 mark]

[Total: 2 marks]

- It is rare for any chemical reaction to have a 100% yield
 - Some product is often lost during a chemical reaction
- Without further information from the question to allow you to calculate the yield, you can only talk qualitatively about yield
 - If a small amount of the product is lost during each step, then more product is lost when there are more steps
 - Therefore, student **B** will lose more products as they have more steps in their preparation of butanoic acid
 - You can think of this crudely:
 - If you start with 10 g of halogenoalkane and lose 1 g with each step
 - Then, student **A** will have $10 - 1 - 1 = 8$ g left
 - While student **B** will have $10 - 1 - 1 - 1 = 7$ g left

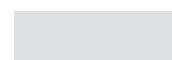
5a

a) The reagents and solvents for reactions 1, 2 and 3 are:

Reaction 1:

- Reagent = NaOH / sodium hydroxide
OR
KOH / potassium hydroxide; [1 mark]
- Solvent = H₂O / water / aqueous; [1 mark]

Reaction 2:



5a

a) The reagents and solvents for reactions 1, 2 and 3 are:

Reaction 1:

- Reagent = NaOH / sodium hydroxide
OR
KOH / potassium hydroxide; [1 mark]
- Solvent = H₂O / water / aqueous; [1 mark]

Reaction 2:

- Reagent = NH₃ / ammonia; [1 mark]
- Solvent = C₂H₅OH / ethanol / alcohol; [1 mark]

Reaction 3:

- Reagent = NaCN / sodium cyanide
OR
KCN / potassium cyanide; [1 mark]
- Solvent = C₂H₅OH / ethanol / alcohol; [1 mark]

[Total: 6 marks]

- There are often marks for reagents, solvents and conditions of organic reactions so make sure you know them

5b

b) The alternative solvents and products for reactions 1 and 3 are:

Reaction 1:

- Alternative solvent = C₂H₅OH / ethanol / alcohol; [1 mark]
- Alternative product = C₂H₄ / ethene; [1 mark]

Reaction 3:

- Alternative solvent = H₂O / water / aqueous; [1 mark]
- Alternative product = C₂H₅OH / ethanol; [1 mark]

[Total: 4 marks]

- This type of question is more commonly asked about reaction 1 but the same principle applies to reaction 3
- You need to know that the use of a solvent can affect a chemical reaction
 - In reaction 1:
 - The use of water as the solvent / sodium hydroxide solution results in the nucleophilic substitution reaction to form an alcohol from the halogenoalkane
 - However, the use of ethanol as the solvent results in the elimination reaction to form an alkene from the halogenoalkane
 - In reaction 2:
 - The use of water as the solvent / sodium hydroxide solution results in the nucleophilic substitution reaction to form a nitrile from the halogenoalkane
 - However, the use of ethanol as the solvent results in the nucleophilic substitution reaction to form an alcohol from the halogenoalkane due to the large number of hydroxide ions

5c

c)

i) The name of the compound CH_3CN is:

- Ethanenitrile; [1 mark]

ii) The type of reaction used to form CH_3CN is:

- Oxidation

OR

Dehydrogenation; [1 mark]

[Total: 2 marks]

- To name CH_3CN :
 - The CH_3C portion of the compound has two carbon atoms with single bonds so its name is ethane based
 - The CN portion of the compound shows that there is a nitrile group
 - So the name is ethanenitrile

- CN , the name is ethanenitrile

- **Careful:** Nitrile compounds are often named incorrectly as the nitrile carbon is missed
- Ethylamine, CH_3NH_2 , has lost four hydrogen atoms to become CH_3CN
 - Loss of hydrogen can be called oxidation or dehydrogenation

5d

d) Comparing the two types of hydrolysis reaction in the production of propanoic acid from the product of reaction 3:

- (The two types of hydrolysis are) acid hydrolysis

AND

Alkali(ne) hydrolysis; [1 mark]

- Acid hydrolysis forms propanoic acid in one step / reaction

AND

Alkali(ne) hydrolysis forms propanoic acid in two steps / reactions; [1 mark]

- Acid hydrolysis forms an ammonium salt / named ammonium salt

AND

Alkali(ne) hydrolysis forms ammonia; [1 mark]

- Both types of hydrolysis require water / aqueous solutions; [1 mark]
- Both types of hydrolysis involve acid

OR

Alkali(ne) hydrolysis also requires acid to acidify the (metal / sodium) carboxylate salt; [1 mark]

[Total: 5 marks]

- You can score the first mark by talking about acid and alkaline hydrolysis in your answer, they do not need to be seen as separate points
- **Remember:** The exam command "compare" means talk about the similarities and differences

